The Role of Diet Containing Seeds of *Elaeis Guineensis* on Heamatological and Serum Biochemical Variables During Intermittent Fasting

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ABSTRACT

This study evaluated the effects of dietary *Elaeis guineensis* seed supplementation, with and without alternate-day intermittent fasting (IF), on anthropometric measurements, haematological indices, and serum biochemical parameters in rats. Thirty-six female rats weighing 80–110g were assigned to six groups: Control, IF, 5% palm kernel seed diet (PKSD), 5% PKSD + IF, 15% PKSD, and 15% PKSD + IF for eight weeks. Anthropometric parameters were measured weekly and blood samples collected at weeks 4 and 8 for haematological and serum liver enzymes estimation. Data were analyzed using one-way analysis of variance followed by Tukey's post hoc test, with statistical significance set at $p<0.05$. The 5% PKSD presented highest body weight gain while 15% PKSD + IF group showed the least weight gain and abdominal circumference. At four weeks, IF significantly increased packed cell volume (PCV) and haemoglobin concentration, while PKSD supplementation increased white blood cell counts, an effect attenuated by concurrent IF. By week eight, IF significantly reduced red blood cell count and PCV, whereas PKSD supplementation maintained significantly higher haemoglobin concentrations. Serum alanine aminotransferase and aspartate aminotransferase activities were elevated in the IF group but were significantly reduced in PKSD-supplemented groups, with the greatest reduction observed in the 5% PKSD + IF group. Dietary supplementation with *Elaeis guineensis* seed ameliorated fasting-induced alterations in haematological indices and reduced elevations in liver enzyme activities, suggesting protective effects on erythropoietic function and hepatic integrity and may serve as functional dietary component for mitigating haematological and hepatic effects associated with alternate-day IF.

Keywords: *Elaeis guineensis*, Intermittent fasting, Hematological variables, Liver enzymes, Palm kernel seed

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1. INTRODUCTION

Intermittent fasting (IF) has emerged as an effective dietary intervention for improving metabolic health and reducing the risk of obesity-related disorders. Unlike continuous calorie restriction, IF involves alternating periods of voluntary fasting and unrestricted feeding, with common approaches including alternate-day fasting, time-restricted eating, and the 5:2 fasting regimen [1]. During fasting, depletion of hepatic glycogen stores initiates a metabolic shift from glucose utilization to fatty acid oxidation and ketogenesis, thereby improving metabolic flexibility. These physiological adaptations have been associated with enhanced insulin sensitivity, reduced adiposity, improved lipid metabolism, and attenuation of systemic inflammation [2]. Consequently, IF has gained considerable attention as a non-pharmacological strategy for the prevention and management of metabolic diseases. Recent meta-analyses have demonstrated that IF significantly reduces body weight, fat mass, fasting insulin concentrations, and glycated haemoglobin (HbA1c), highlighting its therapeutic potential in maintaining metabolic health [3].

In addition to its metabolic benefits, intermittent fasting influences several physiological systems, including

haematopoiesis and liver function. Previous studies have reported alterations in packed cell volume, haemoglobin concentration, erythrocyte count, leukocyte profile, and platelet indices following intermittent fasting, largely due to changes in plasma volume, hormonal regulation, erythropoietic activity, and inflammatory responses [4,5]. Similarly, fasting-induced metabolic adaptations can affect hepatic function through increased fatty acid mobilization and oxidative metabolism, leading to changes in serum biochemical markers such as alanine aminotransferase (ALT) and aspartate aminotransferase (AST). However, these responses are not solely determined by fasting itself but are also influenced by the nutritional quality of the diet consumed during feeding periods. Therefore, identifying dietary components capable of minimizing the adverse haematological and hepatic effects of prolonged fasting remains an important area of nutritional research.

Elaeis guineensis Jacq. (African oil palm), belonging to the family Arecaceae, is an economically important tropical crop widely cultivated across West Africa. While extensive research has focused on palm oil extracted from the fruit mesocarp, comparatively little attention has been given to the nutritional and medicinal potential of the palm kernel seed. Traditionally, palm kernel preparations have been used in the management of convulsions, skin infections, gastrointestinal disorders, and cardiovascular diseases [6,7]. Nutritional analyses have shown that the seed is a rich source of crude fat, predominantly medium-chain fatty acids, and contains appreciable quantities of iron, phosphorus, calcium, B-complex vitamins, and several bioactive phytochemicals, including phenolic compounds and tocotrienols [8]. These nutrients and phytochemicals possess potent antioxidant, anti-inflammatory, and hepatoprotective properties. Tocotrienols have been shown to attenuate hepatic steatosis and oxidative liver injury [9], whereas phenolic compounds suppress inflammatory responses through inhibition of the nuclear factor-kappa B (NF-κB) signalling pathway [10]. In addition, the iron and B-vitamin content of the seed may promote erythropoiesis and haemoglobin synthesis by providing essential substrates required for red blood cell production [11].

Haematological indices are widely recognised as sensitive indicators of nutritional status, erythropoietic function, and immune competence, while serum liver enzymes provide valuable information regarding hepatocellular integrity and metabolic adaptation [12]. Evaluating these biomarkers therefore provides important insight into the physiological consequences of dietary interventions and fasting. Considering the nutritional composition and biological activities of *Elaeis guineensis* seed, it is plausible that dietary supplementation may enhance the adaptive responses to intermittent fasting by supporting erythropoiesis, reducing oxidative stress, and preserving hepatic function.

Although numerous studies have independently demonstrated the beneficial effects of intermittent fasting on metabolic health and the antioxidant and hepatoprotective properties of *Elaeis guineensis*, little is known about the combined effects of palm kernel seed supplementation and alternate-day intermittent fasting on haematological indices and serum biochemical markers. Specifically, it remains unclear whether dietary supplementation with *E. guineensis* seed can mitigate fasting-induced alterations in erythropoietic activity and liver function while preserving the metabolic benefits associated with intermittent fasting. Addressing this knowledge gap is important because identifying functional foods that enhance the physiological safety of intermittent fasting may improve its long-term nutritional and therapeutic application.

Therefore, the present study investigated the effects of dietary supplementation with *Elaeis guineensis* seed, administered with and without alternate-day intermittent fasting, on anthropometric characteristics, haematological

indices, and serum biochemical parameters in female Wistar rats. We hypothesized that *E. guineensis* seed supplementation would attenuate fasting-induced alterations in haematological and hepatic biochemical parameters through its rich nutritional composition and antioxidant phytochemicals, thereby supporting erythropoietic function and maintaining hepatic integrity during alternate-day intermittent fasting.

2. MATERIALS AND METHOD

2.1. Plant material and authentication: Freshly harvested fruits of *Elaeis guineensis* Jacq. were obtained from Igbara-Oke, Ondo State, Nigeria. The fruits were manually cracked to obtain the palm kernel seeds, which were authenticated at the Centre for Research and Development (CERAD), Federal University of Technology, Akure, Nigeria.

2.2 Nutritional analysis of *elaeis guineensis* seed: The palm kernel seeds were thoroughly cleaned, and the shells were removed manually. The kernels were subsequently analyzed for proximate composition, phytochemical constituents, mineral content, and vitamin composition using standard analytical procedures.

2.3 Experimental animals and study design: Thirty-six (36) healthy female Wistar rats weighing between 80 and 110 g were obtained and housed in well-ventilated cages in the Animal House of the Department of Physiology, Federal University of Technology, Akure, Nigeria. The animals were maintained under standard laboratory conditions (temperature: 22–25°C; 12 h light/12 h dark photoperiod) with free access to clean drinking water and standard laboratory chow during a one-week acclimatization period.

Animal care and all experimental procedures were conducted in accordance with the National Institutes of Health (NIH) Guide for the Care and Use of Laboratory Animals [12]. Ethical approval for the study was obtained from the Centre for Research and Development (CERAD), Federal University of Technology, Akure, Nigeria (Approval No. FUTA/CRD/ED/23/134).

Following acclimatization, the animals were randomly assigned into six experimental groups (n = 6 per group) as follows:

Group I (Control): Received standard laboratory chow and water *ad libitum*.

Group II (IF): Subjected to alternate-day intermittent fasting consisting of 24 h fasting followed by 24 h unrestricted access to standard chow and water throughout the study period.

Group III (5% PKSD): Received a diet containing 5% *Elaeis guineensis* seed (Palm Kernel Seed Diet; PKSD) and water *ad libitum*.

Group IV (5% PKSD + IF): Received the 5% PKSD and was simultaneously subjected to alternate-day intermittent fasting, with feed provided only during the feeding periods.

Group V (15% PKSD): Received a diet containing 15% PKSD and water *ad libitum*.

Group VI (15% PKSD + IF): Received the 15% PKSD while undergoing alternate-day intermittent fasting, with feed available only during the feeding periods. All dietary interventions were maintained continuously for eight weeks.

2.4 Experimental diets: Three experimental diets were formulated for this study, and their compositions are presented in Table 1. Rats in the Control and IF groups received standard laboratory chow throughout the experimental period. Animals in Groups III and IV received the diet supplemented with 5% palm kernel seed, whereas Groups V and VI received the diet supplemented with 15% palm kernel seed.

2.5 Body weight determination: Body weights were measured using a calibrated digital weighing balance at baseline and subsequently recorded weekly throughout the eight-week experimental period. Percentage

body weight change was calculated relative to the initial body weight using the following equation:

Percentage Body Weight Change = (Final Body Weight − Initial Body /Weight Initial Body Weight) × 100.

2.6 Anthropometric measurements: Anthropometric measurements were obtained weekly throughout the experimental period. Abdominal circumference was measured using a non-elastic measuring tape according to the method described by Novelli *et al.* [13]. Thoracic circumference and body length (nose-to-anus length) were measured following the murinometric procedures described by Macêdo *et al.* [14].

2.7 Haematological analysis: Blood samples were collected from each animal at Weeks 4 and 8 of the experimental period via retro-orbital venous puncture into ethylenediaminetetraacetic acid (EDTA) anticoagulant tubes. Complete blood count analysis, including packed cell volume (PCV), red blood cell (RBC) count, haemoglobin (Hb) concentration, white blood cell (WBC) count, and platelet count, was performed using an automated hematology analyzer (KT-6200, China).

2.8 Biochemical analysis: Serum was separated by centrifugation and used for the determination of liver function biomarkers. Serum alanine aminotransferase (ALT) and aspartate aminotransferase (AST) activities were determined using commercially available Fortress Diagnostics® assay kits (Fortress Diagnostics, United Kingdom) according to the manufacturer's instructions.

2.9 Statistical analysis: Data are presented as mean ± standard error of the mean (SEM). Statistical analyses were performed using GraphPad Prism version 10.2 (GraphPad Software, San Diego, CA, USA). Differences among multiple groups were analyzed using one-way analysis of variance (ANOVA), followed by Tukey's multiple comparison post hoc test where appropriate. Comparisons between two groups were performed using the unpaired Student's *t*-test. A *p*-value of less than 0.05 ($p < 0.05$) was considered statistically significant.

3. RESULTS

3.1 Nutritional composition of *Elaeis guineensis* seed

The nutritional composition of *Elaeis guineensis* seed is presented in Table 2. Proximate analysis revealed that the seed contained a high crude fat content, with moderate amounts of crude fibre, carbohydrate, ash, and crude protein. The metabolizable energy was 12.6 MJ/kg. Phytochemical analysis demonstrated appreciable concentrations of phenolic acids and flavanols. Mineral analysis showed that phosphorus was the most abundant mineral, followed by calcium, magnesium, iron, potassium, and sodium. Vitamin analysis further revealed the presence of vitamins A, B1, B2, B3, and E.

3.2 Effect of palm kernel seed diets on anthropometric parameters

The effects of palm kernel seed diets on anthropometric parameters during the eight-week intervention are presented in Figure 1. The 5% PKSD group exhibited the greatest progressive increase in body weight throughout the study period. In contrast, the 15% PKSD + IF group recorded the lowest body weight gain. The Control and IF groups demonstrated moderate increases in body weight, whereas the 15% PKSD and both PKSD + IF groups exhibited comparatively lower weight gains.

Head-to-tail length remained relatively stable across all experimental groups throughout the study period. Minor fluctuations were observed in the 5% PKSD and 5% PKSD + IF groups, whereas the 15% PKSD group recorded slightly lower values during the later weeks of the experiment.

Table 1: Feed composition for the experimental diet

	Control (25kg)	5% Kernel (25kg)	15% Kernel (25kg)
Maize	9	7.75	5.25
Wheat offal	5	5	5
Fish meal	2	2	2
Groundnut cake	3	3	3
Soyabean meal	3	3	3
Palm kernel cake	2	0	0
Bone meal	1	1	1
Salt	0.05	0.05	0.05
Premix	0.05	0.05	0.05
Lysine	0.05	0.05	0.05
Methionine	0.05	0.05	0.05
Kernel		1.25	3.75

Table 2: Nutritional composition of palm kernel seed

Content	Value	Content	Value
Metabolic energy (MJ/kg)	12.6	Sodium (mg/100g)	2.14
Ash (%)	6.896	Calcium (mg/100g)	12.14
Crude fat (%)	47. 40	Magnesium (mg/100g)	5.36
Crude fiber (%)	16.329	Iron (mg/100g)	4.24
Crude protein (%)	6.329	Vitamin B1 (mg/g)	0.26770
Carbohydrate (%)	10.358	Vitamin B3 (mg/g)	1.10476
Phenolic acids (mg/100g)	143.64	Vitamin E (mg/g)	2.036328
Flavanols (mg/100mg)	6.604	Vitamin A [mcg (RAE)]	56.056
Phosphorus (mg/100g)	33.47	Vitamin B2 (mg/g)	0.012186
Potassium (mg/100mg)	3.85		

Thoracic circumference exhibited minor fluctuations across all groups. The Control and IF groups maintained relatively stable measurements, while the 5% PKSD group showed a slight increase during the early phase of the study followed by stabilization. Greater variability was observed in the 15% PKSD and 15% PKSD + IF groups.

Abdominal circumference also varied throughout the study period. Moderate fluctuations were observed in the Control and IF groups, whereas intermittent increases were recorded in the 5% PKSD group. Lower abdominal circumference values were observed in the 15% PKSD and, particularly, the 15% PKSD + IF groups

3.3 Effect of palm kernel seed diets on haematological parameters after 4 weeks of intervention

Figure 2 presents the effect of PKSD on haematological indices after 4 weeks of intermittent fasting. The result revealed a significant increase in PCV in the IF, 5%

PKSD, 15% PKSD and 15% PKSD groups compared to the control group ($p < 0.05$). The 5% PKSD group showed a significant increase when compared with 5% PKSD + IF group.

The 5% PKSD and 5% PKSD + IF groups showed a significantly reduced RBC counts when compared with control and 15% PKSD + IF groups Haemoglobin concentration was significant increase in the IF, 5% PKSD, 15% PKSD and 15% PKSD groups compared to the control ($p < 0.05$).

The 5% PKSD, 15% PKSD and 15% PKSD + IF groups showed significant increase in WBC counts when compare to the control and 5% PKSD + IF group. However, the 5% PKSD + IF group exhibited a significant reduction in WBC counts compared to 5% PKSD, 15% PKSD, and 15% PKSD groups ($p < 0.05$).

The IF, 5% PKSD +IF, 15% PKSD and 15% PKSD groups showed significant increase in platelet counts when compared in the control and 5% PKSD groups. Conversely, the 5% PKSD group revealed a significant decrease in platelet counts when compared to IF, 5% PKSD + IF, 15% PKSD and 15% PKSD + IF groups

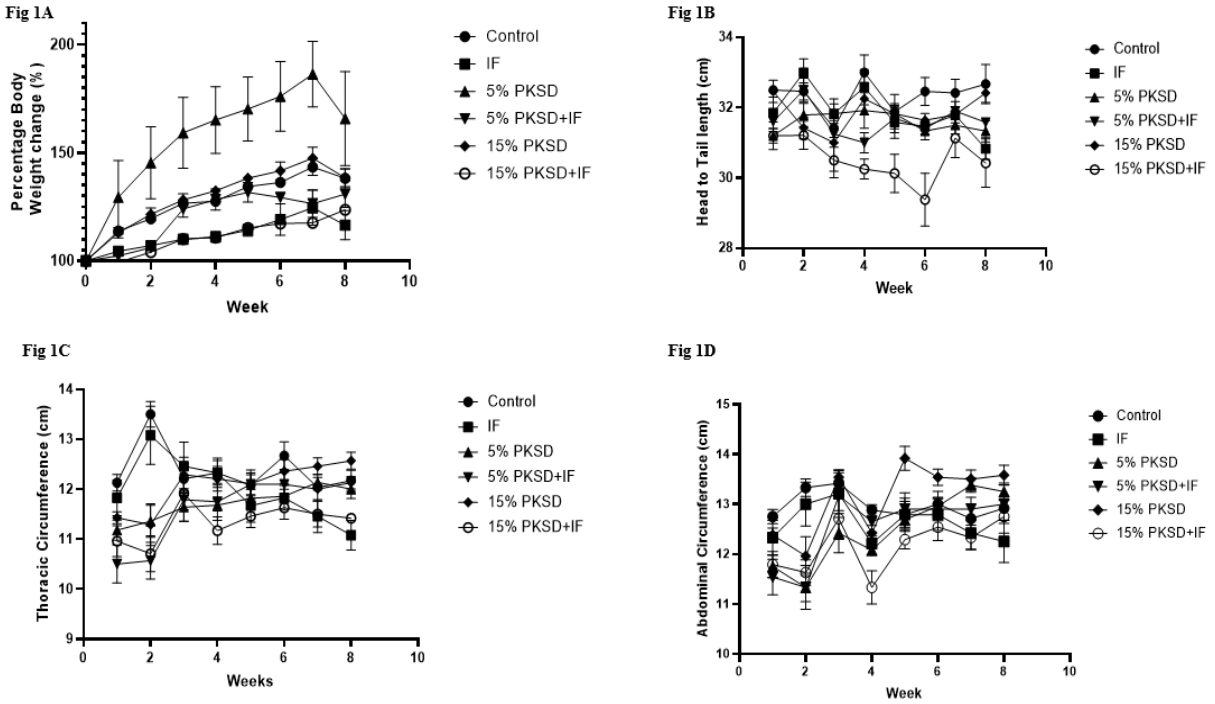


Figure 1: Effect of PKSD on anthropometry variables (a. percentage body weight change, b. Head to tail length, c. Thoracic Circumference and d. Abdominal Circumference) following 8 weeks of intermittent fasting.

Values are expressed as Mean ± SEM. N=5. a,b,c,d,e and f represent significantly different when compared with control, IF, 5% PKSD, 5% PKSD + IF, 15% PKSD, 15% PKSD + IF respectively. IF= Intermittent fasting; PKSD= Palm Kernel Seed Diet.

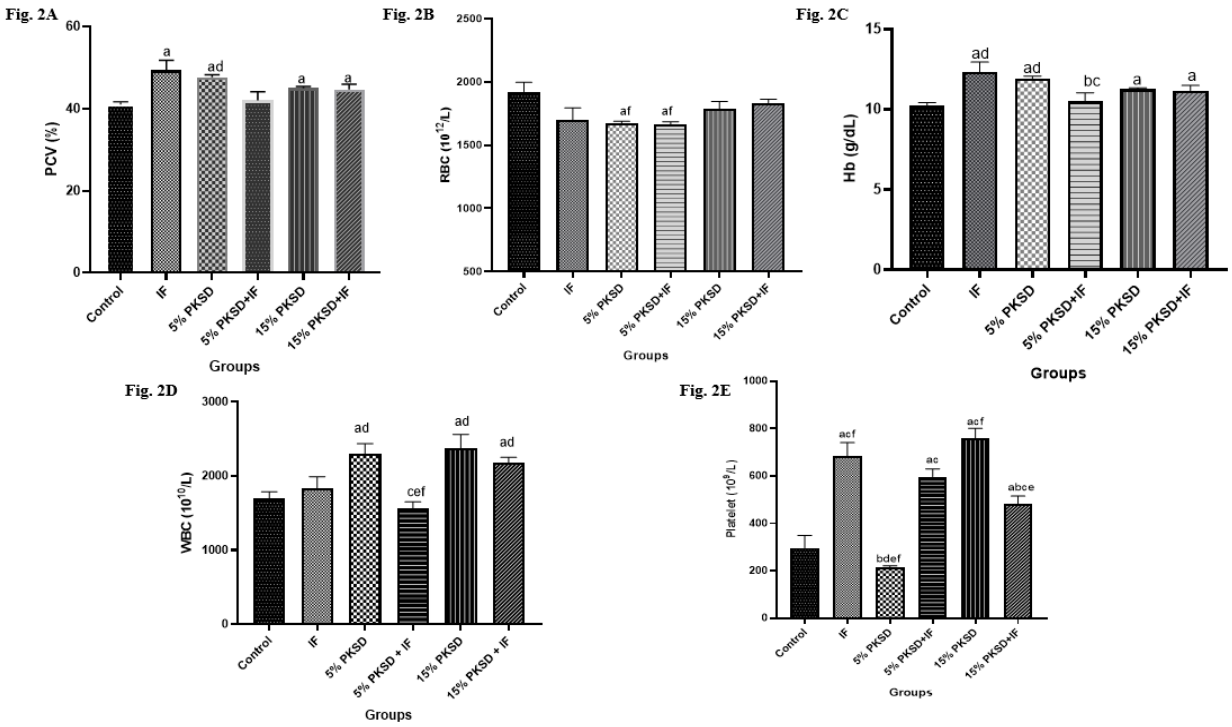


Figure 2: Effect of PKSD on hematological indices after 4 weeks of intermittent fasting.
Values are expressed as Mean ± SEM. N=5. a,b,c,d,e and f represent significantly different when compared with control, IF, 5% PKSD, 5% PKSD + IF, 15% PKSD, 15% PKSD + IF respectively. IF= Intermittent fasting; PKSD= Palm Kernel Seed Diet.

3.4 Effect of Palm Kernel Seed Diets on Haematological Parameters After Eight Weeks of Intervention

The effects of palm kernel seed diets on haematological indices after eight weeks of intervention are presented in Figure 3. Packed cell volume was significantly lower in the 5% PKSD and 5% PKSD + IF groups than in the IF group ($p < 0.05$). Conversely, the IF group had significantly lower PCV than the Control, 15% PKSD, and 15% PKSD + IF groups ($p < 0.05$).

Red blood cell counts were significantly lower in the IF, 5% PKSD, 5% PKSD + IF, and 15% PKSD + IF groups than in the Control and 15% PKSD groups ($p < 0.05$). Among all experimental groups, the IF group recorded the lowest RBC count.

Haemoglobin concentration was significantly lower in the IF and 5% PKSD + IF groups than in the Control, 5% PKSD, and 15% PKSD groups ($p < 0.05$). The 5% PKSD and 15% PKSD groups exhibited significantly higher haemoglobin concentrations than the IF and 5% PKSD + IF groups ($p < 0.05$).

White blood cell counts were significantly lower in the IF, 5% PKSD, and 15% PKSD + IF groups than in the Control group ($p < 0.05$). The 5% PKSD group recorded the lowest WBC count among all experimental groups. In addition, the 15% PKSD + IF group had significantly lower WBC counts than the Control, IF, and 5% PKSD + IF groups ($p < 0.05$).

Platelet counts were significantly higher in the 5% PKSD, 5% PKSD + IF, and 15% PKSD groups than in the Control and 15% PKSD + IF

groups ($p < 0.05$). The 15% PKSD + IF group recorded the lowest platelet count among the treatment groups.

3.5 Effect of Palm Kernel Seed Diets on Liver Enzyme Activities

The effects of palm kernel seed diets on serum liver enzyme activities after eight weeks of intervention are presented in Figure 4. Alanine aminotransferase (ALT) activity was significantly lower in the 5% PKSD + IF group than in the Control group ($p < 0.05$). In addition, ALT activity was significantly lower in the 5% PKSD, 5% PKSD + IF, and 15% PKSD groups than in the IF group ($p < 0.05$). Aspartate aminotransferase (AST) activity was significantly lower in the 5% PKSD + IF and 15% PKSD + IF groups than in the Control group ($p < 0.05$).

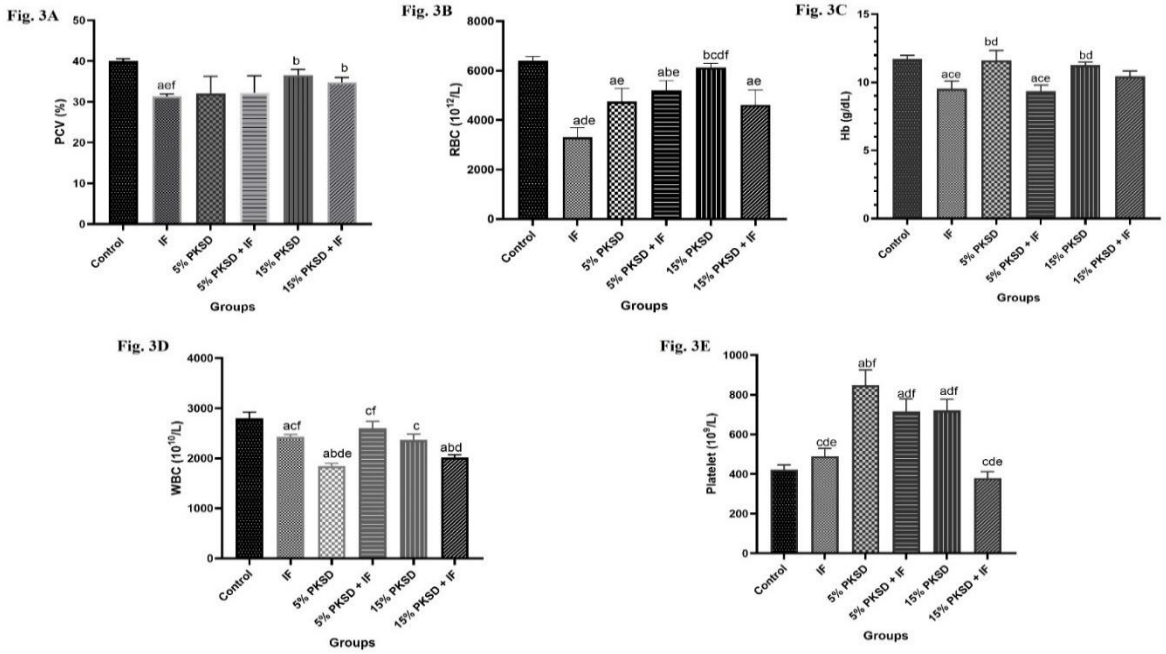


Figure 3: Effect of PKSD on haematological indices after 8 weeks of intermittent fasting.
Values are expressed as Mean ± SEM. N=5. a,b,c,d,e and f represent significantly different when compared with control, IF, 5% PKSD, 5% PKSD + IF, 15% PKSD, 15% PKSD + IF respectively. IF= Intermittent fasting; PKSD= Palm Kernel Seed Diet.

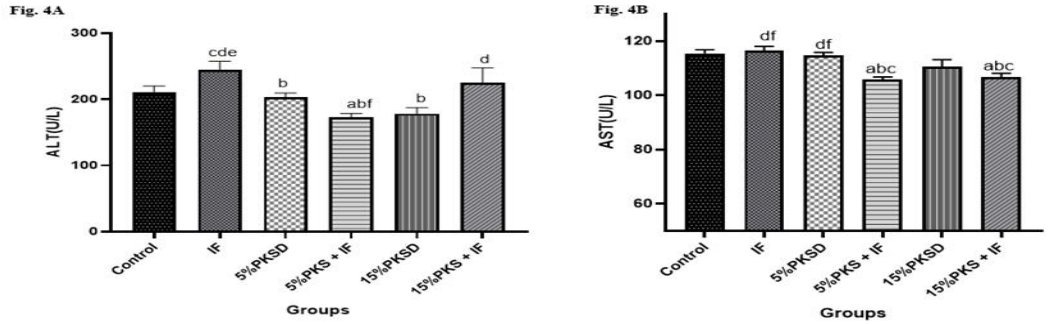


Figure 4 A and B; Alanine amino transferase (ALT) and Aspartate amino transferase (AST) respectively.
Values are expressed as Mean ± SEM. N=5. a,b,c,d,e and f represent significantly different when compared with control, IF, 5% PKSD, 5% PKSD + IF, 15% PKSD, 15% PKSD + IF respectively. IF= Intermittent fasting; PKSD= Palm Kernel Seed Diet.

DISCUSSION

The present study investigated the effects of dietary supplementation with *Elaeis guineensis* seed, with and without alternate-day intermittent fasting, on anthropometric characteristics, haematological indices, and serum biochemical parameters in female Wistar rats. Nutritional analysis demonstrated that the palm kernel seed contained high crude fat, moderate amounts of crude fibre, carbohydrate, crude protein, and ash, with a metabolizable energy value of 12.6 MJ/kg. The seed also contained appreciable concentrations of phenolic acids, flavanols, minerals, and vitamins. These findings are consistent with previous reports describing the nutritional composition and functional properties of palm kernel products [15,16]. The abundance of medium-chain fatty acids and antioxidant phytochemicals supports the nutritional value of *E. guineensis* seed and its potential as a functional dietary ingredient [17–19].

The anthropometric findings showed that rats fed the 5% PKSD exhibited the greatest body weight gain, whereas the combination of 15% PKSD and intermittent fasting produced the lowest weight gain and reduced abdominal circumference. These observations are consistent with reports that intermittent fasting promotes weight reduction by enhancing lipolysis and increasing fatty acid oxidation during periods of energy restriction [20–22]. The relatively greater weight gain observed in the 5% PKSD group may reflect the higher energy density of the supplemented diet. Meanwhile, the absence of marked differences in body length among the groups suggests that neither intermittent fasting nor palm kernel seed supplementation adversely affected skeletal growth during the experimental period, consistent with previous studies evaluating short-term dietary interventions [23,24].

The haematological findings demonstrated that intermittent fasting produced time-dependent effects on erythrocyte indices. After four weeks, packed cell volume and haemoglobin concentration were significantly higher in the fasting group than in the control group, findings that agree with previous reports attributing early increases in these parameters to haemoconcentration associated with reduced plasma volume during fasting [4]. Experimental studies have also suggested that short-term caloric restriction may transiently stimulate erythropoiesis [25]. However, by the eighth week, prolonged intermittent fasting was associated with lower red blood cell count, packed cell volume, and haemoglobin concentration, indicating that sustained fasting may adversely affect erythropoietic function. Similar observations have been reported in prolonged calorie-restriction models, where reduced erythropoietin activity and altered iron metabolism have been associated with diminished red blood cell production [25].

Palm kernel seed supplementation appeared to moderate some of these haematological changes. Animals receiving the 5% and 15% PKSD diets maintained higher haemoglobin concentrations than those subjected to intermittent fasting alone after eight weeks. This finding may be related to the nutritional composition of the seed, particularly its iron and B-vitamin contents, which are essential for erythropoiesis and haemoglobin synthesis [11]. However, further mechanistic studies are required to establish the specific pathways responsible for these observations.

Alterations in leukocyte counts also varied according to dietary treatment and duration of exposure. The higher white blood cell counts observed in some PKSD-supplemented groups after four weeks may reflect an early physiological response to dietary modification, consistent with previous reports linking high-fat diets with modest increases in circulating leukocyte counts [18]. By contrast, lower leukocyte counts observed after prolonged intermittent fasting agree with studies suggesting that fasting may reduce inflammatory activity through modulation of inflammatory mediators [18]. Because inflammatory biomarkers were not measured

in the present study, these explanations should be interpreted cautiously.

Platelet counts also differed among the treatment groups following prolonged intervention. Although the mechanisms underlying these changes remain unclear, previous experimental evidence suggests that fasting and alterations in energy metabolism may influence thrombopoiesis [25]. Additional studies are required to determine the mechanisms responsible for these observations.

Serum liver enzyme activities further demonstrated the influence of dietary intervention on hepatic function. Rats subjected to intermittent fasting alone exhibited higher ALT activity than several PKSD-treated groups after eight weeks. Similar increases in transaminase activity have been reported during prolonged fasting and have been attributed to increased fatty acid mobilization and hepatic metabolic adaptation [27]. Conversely, PKSD supplementation was associated with lower ALT activity, particularly when combined with intermittent fasting. This observation is consistent with previous studies reporting antioxidant and hepatoprotective properties of palm-derived phenolic compounds and tocotrienols, which may reduce oxidative stress and lipid peroxidation in hepatic tissue [15,28,29]. Nevertheless, the present study did not directly assess oxidative stress biomarkers; therefore, these proposed mechanisms remain speculative.

Similarly, AST activity was lower in the combined PKSD and intermittent fasting groups than in the control group. Although reductions in serum AST activity may suggest improved hepatic integrity, AST is also present in skeletal and cardiac muscle. Consequently, interpretation of AST changes alone should be made cautiously and in conjunction with other biochemical and histological assessments [30].

Overall, the findings indicate that dietary supplementation with *Elaeis guineensis* seed may influence haematological and hepatic biochemical responses during alternate-day intermittent fasting. The beneficial effects observed may be associated with the nutritional and antioxidant constituents of the seed; however, additional studies incorporating oxidative stress biomarkers, inflammatory mediators, erythropoietic regulators, and histopathological evaluation are required to clarify the underlying mechanisms.

5. CONCLUSION

The present study demonstrates that dietary supplementation with *Elaeis guineensis* seed influences haematological indices and serum liver enzyme activities in female Wistar rats, with the magnitude of these effects varying according to the level of supplementation and the presence of alternate-day intermittent fasting. While intermittent fasting alone produced an initial increase in haematological indices after four weeks, prolonged fasting was associated with reductions in erythrocyte-related parameters after eight weeks. In contrast, palm kernel seed supplementation was associated with the maintenance of haemoglobin concentration and lower serum ALT and AST activities compared with intermittent fasting alone.

These findings suggest that *Elaeis guineensis* seed may help attenuate some haematological and hepatic alterations associated with prolonged alternate-day intermittent fasting, possibly because of its favourable nutritional composition and antioxidant constituents. However, the mechanisms underlying these effects were not directly investigated in the present study and warrant further research.
Overall, *Elaeis guineensis* seed shows potential as a functional dietary ingredient capable of supporting physiological adaptation during alternate-day intermittent fasting. Further studies, including mechanistic investigations and clinical trials, are required to confirm these findings and determine their relevance to human health.

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